

Task 1 - QA & Debug Report

Candidate	Uday Upadhyay
Role	Automation & QA Developer (take-home assessment)
App tested	RealWorld "Conduit" (Angular frontend demo) - https://demo.realworld.show/
Test date	30 Sep 2026
Environment	Windows, Microsoft Edge, DevTools + Lighthouse (Mobile, Accessibility)

Summary

I picked the public RealWorld demo (a Medium-style blogging app) and ran the main flows: sign-up, login, create article, edit/delete, comment and logout, then tried bad input (empty fields, invalid email, very short password, over-long tags, HTML in the article body) and ran a Lighthouse accessibility audit. I found **8 issues**: 3 High, 3 Medium, 2 Low. The most important theme is that the app **does not validate email and password**, so accounts can be created with an email like "ab" and a 2-character password. Note: the link suggested in the brief, demo.realworld.io, returned "404 NoSuchBucket" (site is down), so I used demo.realworld.show instead.

1. Bug table

#	Title / Summary	Steps to Reproduce	Expected vs Actual	Severity	Suspected Cause
1	Invalid email accepted at sign-up	1. Open Sign up 2. Username bugtest1 , Email ab , Password 22 3. Click Sign up 4. Open Settings	Expected: "Enter a valid email" error and no account. Actual: Account created, user logged in, and Settings shows email "ab" saved. (Fig. 1, 2)	High	No email-format validation on the form or on the register API.
2	2-character password accepted	1. Open Sign up 2. Fill Username bugtest1 , Email ab , Password 22 3. Click Sign up	Expected: Minimum length / strength error. Actual: Account created with a 2-character password. (Fig. 1)	High	No password policy (min length / complexity) in the frontend or backend.
3	Invalid email accepted and saved in Settings	1. Login as bugtest1 2. Settings, change email to xyz 3. Click Update Settings 4. Refresh the page and reopen Settings	Expected: Validation error; old email kept. Actual: No error; after refresh the email field showed "xyz". (Fig. 3 shows the form before submitting)	High	Same missing email validation, this time on the update-user API.
4	Raw technical error for invalid tag	1. Login 2. New Article: Title/Description/Body = "text" 3. Enter tag <script>alert(1)</script> (25 chars) 4. Click Publish Article	Expected: Friendly message, e.g. "Each tag must be under 20 characters". Actual: "body tagList is an optional list of less than 10 strings of less than 20 chars" (internal field names). (Fig. 4)	Low	Backend validation message shown to the user as-is, with no frontend translation.
5	Images have no alt text (Lighthouse)	1. Open Home page 2. DevTools, Lighthouse, Accessibility, Mobile, Analyze	Expected: Every meaningful image has an alt attribute. Actual: "Image elements do not have [alt] attributes" (Fig. 5, 6)	Medium	Avatar and other images rendered without alt.
6	Links without a discernible name (Lighthouse)	Same audit as #5	Expected: Each link has visible text or an aria-label. Actual: "Links do not have a discernible name" - screen readers cannot say where the link goes. (Fig. 6)	Medium	Icon or image-only links without text / aria-label.
7	Low colour contrast (Lighthouse)	Same audit as #5	Expected: WCAG AA contrast (4.5:1 for body text). Actual: "Background and foreground colors do not have a sufficient contrast ratio"; e.g. light-grey nav links and "Read more..." are faint. (Fig. 6)	Medium	Light-grey text on white in the CSS theme.

#	Title / Summary	Steps to Reproduce	Expected vs Actual	Severity	Suspected Cause
8	Missing lang attribute on <html> (Lighthouse)	Same audit as #5	Expected: <html lang="en">. Actual: "<html> element does not have a [lang] attribute". Overall Accessibility score: 67/100 . (Fig. 5, 6)	Low	lang not set in the index.html template.

2. Root-cause analysis: Bug #1 (invalid email accepted)

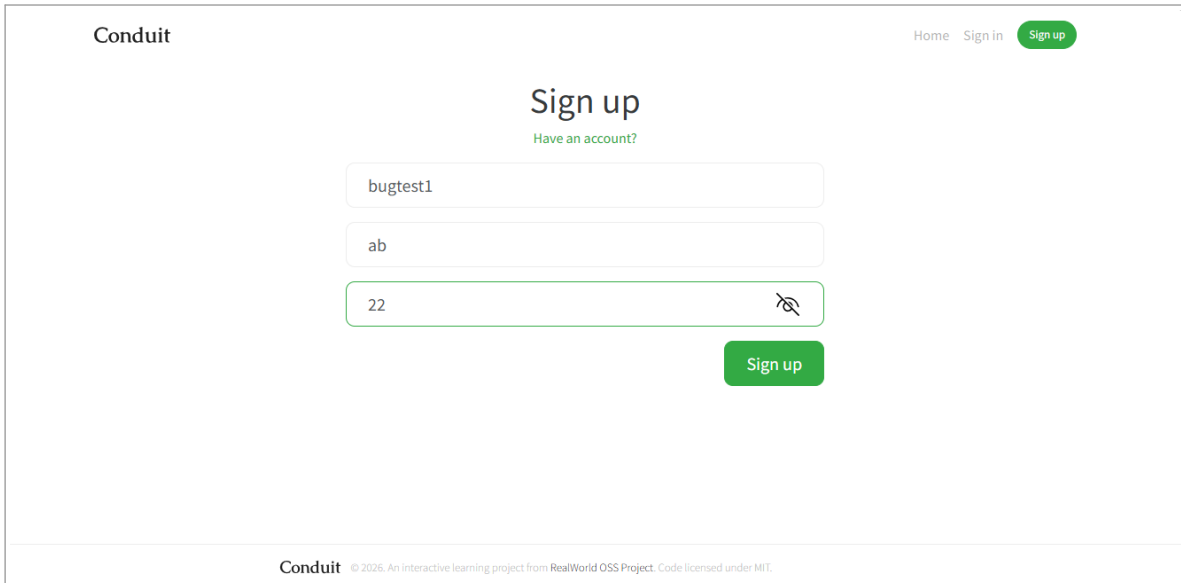
What is happening: The sign-up form accepts the email "ab", creates the account, logs the user in, and the same invalid value is stored and later shown in Settings. **Why:** An email address has to be validated in two places: in the browser form (for fast feedback) and on the server API (because the browser check can be bypassed). Here neither rejected the value, and the API returned a success response and saved it. I did not have access to the source code, so this is inferred from behaviour: the same invalid-email problem also happens on the Settings update (Bug #3), which suggests one missing server-side rule shared by the register and update-user endpoints, not a one-off form glitch. The server does have a validation layer (empty article fields and over-long tags are rejected with HTTP 422), so the email rule looks simply missing. **Impact:** password-reset and notification emails cannot be delivered, junk accounts can be created, and data quality suffers. **How I would fix it:** add a frontend validator (input type="email" plus Angular's email validator) with an inline message; add server-side format validation on both the register and update-user endpoints that returns 422 with a clear message; and add automated tests for both endpoints with invalid emails like "ab" and "xyz". The same change should add a minimum password length (Bug #2), ideally 8 characters.

3. Things I tested that worked correctly (not counted as bugs)

- Logout, then browser Back: the app shows the sign-in/sign-up page and does not expose the logged-in page.
- Publishing an empty article: rejected with clear messages ("title can't be blank", etc.). The 3 errors of HTTP 422 in the DevTools console came from these tests, so they are expected.
- Double-clicking Publish: only one article was created (no duplicate).
- Article body with <script>alert(1)</script>: no popup appeared; the script text was removed from the body.

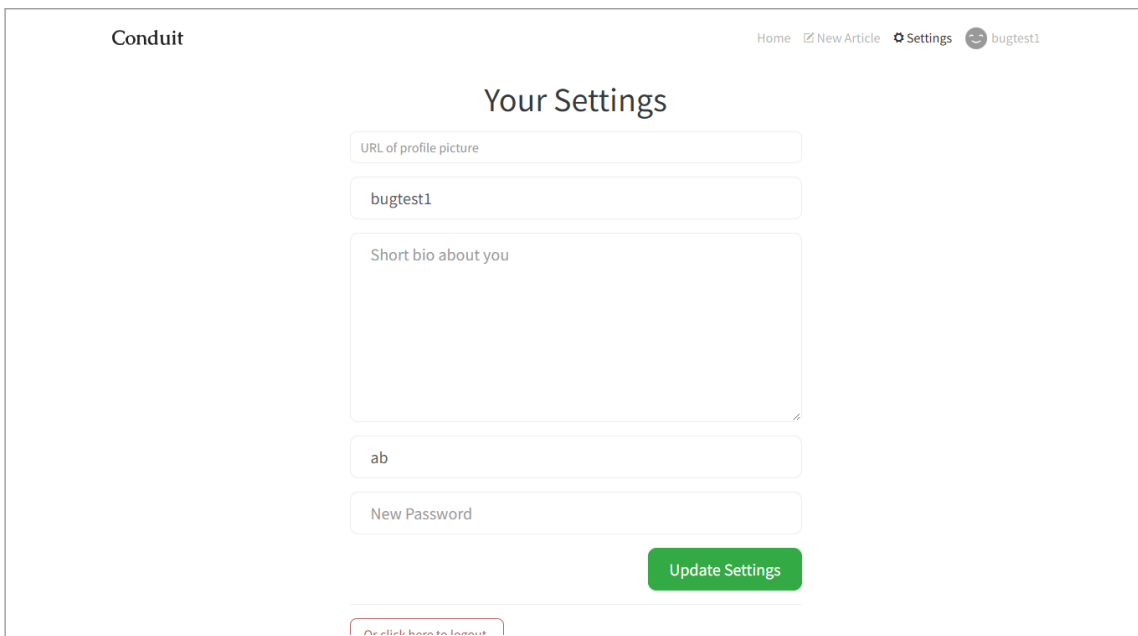
Follow-up not confirmed: a body of showed a broken-image icon and no popup. I did not verify in the DOM inspector what was rendered, so I have not counted it as a bug; it is worth a proper check of how the app sanitises HTML in article bodies.

4. Evidence (screenshots)



The screenshot shows the 'Sign up' page of the Conduit application. The header includes the 'Conduit' logo, links for 'Home', 'Sign in', and a green 'Sign up' button. The main heading is 'Sign up' with a link 'Have an account?'. The form contains three input fields: 'Username' with the value 'bugtest1', 'Email' with the value 'ab', and 'Password' with the value '22'. A green 'Sign up' button is positioned below the fields. The footer contains the text 'Conduit © 2026. An interactive learning project from RealWorld OSS Project. Code licensed under MIT.'

Fig. 1 - Sign-up form filled with Username bugtest1, Email "ab", Password "22" before clicking Sign up (Bugs #1, #2).



The screenshot shows the 'Your Settings' page of the Conduit application. The header includes the 'Conduit' logo, links for 'Home', 'New Article', 'Settings', and a user profile icon labeled 'bugtest1'. The main heading is 'Your Settings'. The form contains five input fields: 'URL of profile picture', 'Username' with the value 'bugtest1', 'Short bio about you', 'Email' with the value 'ab', and 'New Password'. A green 'Update Settings' button is positioned below the fields. The footer contains the text 'Or click here to logout'.

Fig. 2 - After sign-up: user bugtest1 is logged in and Settings shows the invalid email "ab" saved (Bug #1).

Conduit

HomeNew ArticleSettingsbugtest1

Your Settings

URL of profile picture

bugtest1

Short bio about you

xyz

New Password

Fig. 3 - Settings with email changed to "xyz" before clicking Update Settings. After updating and refreshing, the field showed "xyz" (Bug #3).

Conduit

HomeNew ArticleSettingsbugtest1

body tagList is an optional list of less than 10 strings of less than 20 chars

text

text

text

Enter tags

<script>alert(1)</script>

Publish Article

Fig. 4 - Raw backend message shown for an over-long tag (Bug #4).

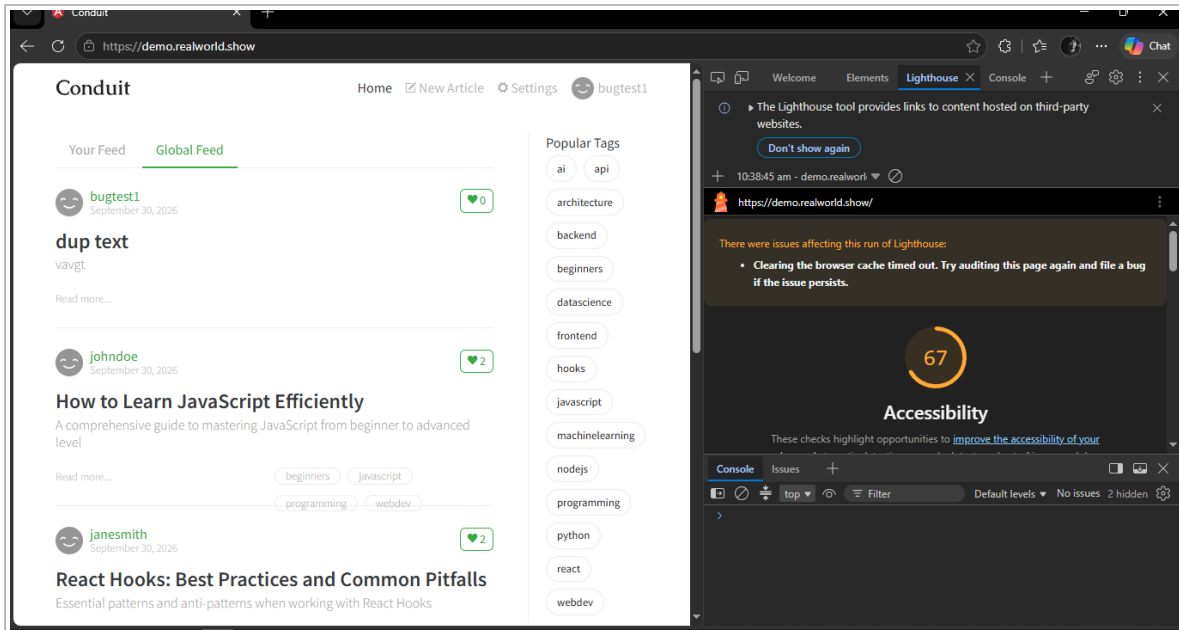


Fig. 5 - Lighthouse (Mobile, Navigation) on the Home page: Accessibility score 67 (Bugs #5-#8).

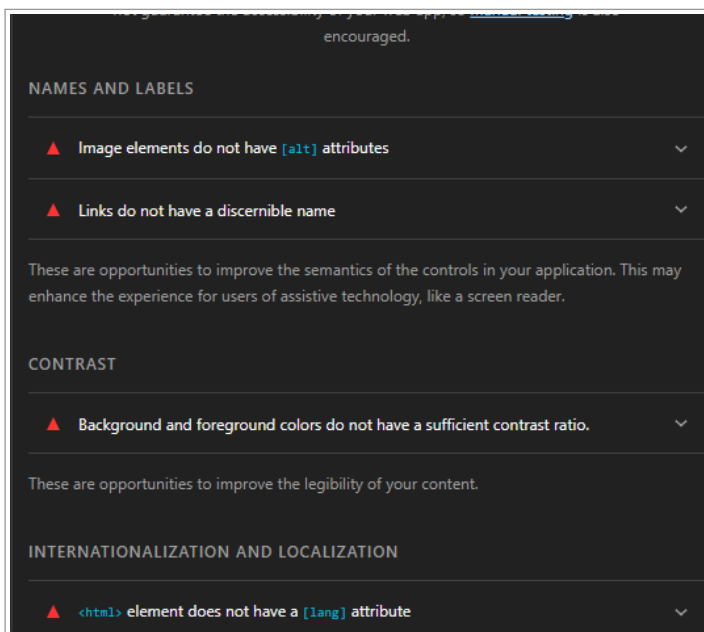


Fig. 6 - Lighthouse findings: missing alt attributes, links without a discernible name, insufficient contrast, missing lang attribute (Bugs #5-#8).